

Magnitude and Phase Aware Spectral Convolutional Neural Networks: Explainable Frequency-Domain Classification

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Abstract. Convolutional neural networks (CNNs) have demonstrated strong performance in visual recognition tasks. However, conventional CNN architectures primarily operate on spatial-domain inputs, which may overlook important discriminative patterns and consequently limit model performance. In medical imaging, CNN models can also process frequency-domain representations obtained through the fast Fourier transform (FFT), which capture informative high-frequency features that are often less apparent in the spatial domain. Despite this advantage, interpretability in the frequency domain remains challenging because identifying the regions of attention in the original image space is non-trivial. In medical imaging applications, where transparent and trustworthy decision support is critical, this lack of explainability restricts clinical adoption. To address this challenge, this paper presents a spectral learning framework with explainability for interpreting frequency-domain CNN predictions. The input red, green, and blue (RGB) image channels are transformed using FFT to construct a 9-channel representation consisting of magnitude, cosine phase, and sine phase components. Although phase information provides localization-related cues, it is highly sensitive to perturbations and may introduce noise that degrades classification performance. To mitigate this issue, first-order gradients of the activation maps are fused with the activation maps before global average pooling. This integration stabilizes learning and enhances the recovery of discriminative spectral signatures. The proposed methodology is evaluated on both medical and non-medical image datasets using multiple CNN architectures to validate its robustness. Qualitative and quantitative results demonstrate the effectiveness of the proposed approach. Furthermore, the framework provides explainability for frequency-domain CNN

models, making it particularly suitable for trustworthy medical imaging applications.

Keywords: Convolutional Neural Networks (CNNs) · Frequency-domain learning · Fast Fourier Transform (FFT) · Explainability · Class Activation Mapping (CAM) · Spectral Feature Representation · Spatial-Frequency Localisation

1 Introduction

Convolutional neural networks (CNNs) have significantly improved image classification performance across various tasks [6]. Models like ResNet-50 typically operate on spatial-domain inputs, analyzing raw pixel intensities [42]. However, spatial representations may miss important discriminative details [39]. In contrast, frequency-domain representations using the fast Fourier transform (FFT) capture high-frequency structural and textural patterns that are often overlooked by standard CNNs [11]. This is especially important in medical imaging, where subtle structural and texture variations are crucial for diagnosis. An illustration of spatial and frequency-domain representations is shown in Fig. 1.

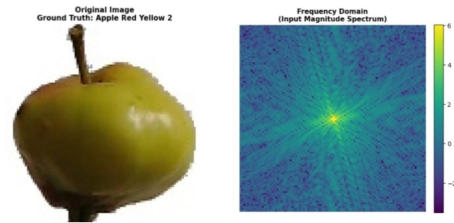


Fig. 1: **Illustration of spatial and frequency domain representations of an image.** The left panel shows the spatial domain image, while the right panel shows its corresponding frequency-domain representation obtained by using FFT.

Frequency-domain learning strengthens medical image analysis by enriching feature representation and retaining fine diagnostic detail, yet models trained on spectra are hard to interpret: frequency components have no direct spatial correspondence, so the image regions driving a prediction remain unclear [39,42], a serious concern in medical imaging, where transparent and trustworthy decision support is essential [16,24]. Two challenges follow. First, spectral inputs carry discriminative magnitude and phase information that spatial CNNs under-utilise, but feeding raw Fourier spectra to a network hurts accuracy because phase is highly sensitive to perturbations. Second, CAM-style explanation methods implicitly assume spatial inputs, so a spectrally trained model cannot be explained

in a form a clinician can verify. This work bridges both within a unified framework: a frequency-aware class activation mapping procedure identifies salient spectral contributions and maps them back to the spatial domain, yielding class-discriminative attribution maps that localise diagnostically relevant regions with fewer visual artefacts [32].

The main contributions of this work are summarised as follows:

1. A magnitude and phase-aware spectral input representation, in which each RGB channel is decomposed via the FFT into log-magnitude, cosine-phase, and sine-phase components, yielding a 9-channel encoding that preserves both spectral energy and localisation cues while remaining continuous and stable for learning.
2. A dual-frequency feature fusion (DFFF) module that applies a second-level FFT to the deepest activation maps and adaptively combines spatial and spectral features through channel attention and learnable fusion weights, mitigating the adverse effects of noise introduced by explicit phase encoding.
3. A frequency-domain explainability mechanism: CAM-style attributions are computed directly on the fused spectral features and mapped back to the spatial domain through the inverse FFT with preserved phase, producing pixel-level saliency maps for a network that never sees raw pixels.
4. An extensive evaluation across five CNN backbones on one medical (Brain Tumor MRI) and one non-medical (Fruits-360) dataset, demonstrating that the framework generalises across architectures and domains while providing interpretable evidence for its decisions.

2 Related Work

Early work showed that networks can learn directly from spectral inputs: CNNs trained on JPEG DCT coefficients gained speed without losing accuracy [12], DFT/DCT features paired with residual CNNs aided breast cancer histopathology [22], and FDCNN used RFFT-based layers with channel-independent convolution to cut complexity while keeping fine detail [11]. Xu et al. exposed a CNN bias toward low-frequency DCT components [42], formalised by Rahaman et al., who showed networks fit low-frequency functions first [26]; Totterstrom found CNNs perform worse on raw Fourier inputs than on RGB [39], while tKFC-Net embedded twin-kernel Fourier convolution into U-Net for sharper boundaries [35] (see [19] for a survey). More recent designs treat spectral operations as core architectural components: FcaNet generalised channel attention via the DCT [25], GFNet replaced self-attention with learnable global Fourier filters [28], and was later extended to adaptive Fourier neural operators [13] and adaptive frequency filters for token mixing [7]. Wavelet-integrated downsampling improves noise robustness [18], FreqFusion brings frequency-aware fusion to dense prediction [5], and frequency cues aid generalisable deepfake detection [3]. In medical imaging, deformable large-kernel attention [2] and vision transformers [1] keep pushing accuracy. Collectively, these works confirm that explicit spectral

modelling strengthens representations, yet they target accuracy and efficiency, leaving frequency-domain explainability unaddressed.

On the explainability side, XAI seeks to make model predictions understandable [16]. The dominant model-agnostic methods, LIME, which fits a local interpretable surrogate via superpixel perturbations, and SHAP, which attributes importance through game-theoretic Shapley values, both struggle with images: LIME localises imprecisely and is expensive. At the same time, SHAP scales poorly to high-dimensional inputs, and its feature-independence assumption weakens attribution reliability [29,16,20]. CAM methods avoid these issues by operating directly on convolutional feature maps, giving precise localisation from a single forward and partial backward pass without architectural changes [32,4]. The family keeps evolving: LayerCAM aggregates shallow and deep activations for finer localisation [17], van der Velden et al. note the dominance of spatial saliency maps across medical XAI [40], and Enguita et al. combined Haar wavelet decomposition with Grad-CAM as a first step toward spectral attribution [9] – though on a spatially trained model rather than a network consuming Fourier-domain inputs. Recent efficiency-focused designs push CAM-style interpretability further, from low-rank formulations for medical analysis [38] and aggregated architectures for interpretable radiograph classification [37] to lightweight, intrinsically explainable transformers for clinical video on edge devices [36].

Positioning this work against the state of the art reveals three concrete research gaps. *Gap 1 – the phase is discarded or mishandled:* current spectral architectures such as FcaNet [25], GFNet [28], and adaptive frequency filters [7] operate on magnitude-like quantities or learned filters and do not exploit the localisation information carried by the Fourier phase, while the direct use of raw phase is known to harm accuracy [39]. *Gap 2 – single-level spectral views:* existing methods apply frequency analysis either to the input [42] [11] or inside isolated layers [35] [5]. To the best of our knowledge, existing methods do not jointly combine an explicit spectral input encoding with a second spectral analysis of the deepest features, leaving cross-domain complementarity unexploited. *Gap 3 – lack of spectral explainability:* Most CAM-style attribution methods, from Grad-CAM [32] to LayerCAM [17], assume spatial inputs and activations, so instance-level explanations for networks operating directly on Fourier-domain representations have received limited attention in the literature. This work addresses these gaps through a stable magnitude and sine-cosine phase encoding (Gap 1), a dual-frequency feature fusion module linking input-level and feature-level spectra (Gap 2), and a CAM procedure that identifies decisive frequencies and maps them back to pixel regions through the inverse FFT (Gap 3).

3 Methodology

The proposed framework transforms spatial-domain images into the frequency domain using the FFT, which decomposes each input into magnitude and phase components. The phase encodes critical spatial localisation information but is inherently sensitive to noise and discontinuities. To address this, each RGB chan-

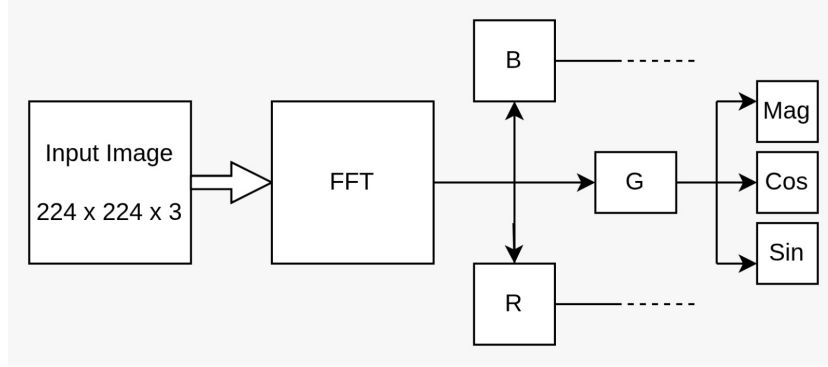


Fig. 2: Each RGB channel is transformed into the frequency domain and decomposed into log-magnitude, cosine-phase, and sine-phase components (illustrated for the G channel), forming a nine-channel spectral representation that serves as input to the CNN for improved feature robustness and localisation.

nel is represented using log-magnitude along with sine and cosine transformations of the phase, ensuring a stable and continuous encoding. This results in a nine-channel representation (3 channels $\times$ magnitude, $\cos \phi$, $\sin \phi$), effectively preserving both spectral characteristics and structural details for improved feature learning. The overall proposed system diagram is shown in Figure 2.

3.1 Spatial-to-frequency domain conversion

The 2D discrete Fourier transform is applied to each colour channel independently and the resulting complex spectrum is centred with `fftshift` before being split into magnitude $M_c(u, v)$, which reflects texture and edge content, and phase $\Phi_c(u, v)$, which retains the spatial arrangement:

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad (1)$$

where u and v are spatial frequency coordinates and $F(u, v)$ is complex-valued with real (Re) and imaginary (Im) parts [6,31]:

$$M_c(u, v) = \sqrt{Re(F(u, v))^2 + Im(F(u, v))^2} \quad (2)$$

$$\Phi_c(u, v) = \arctan \left(\frac{Im(F(u, v))}{Re(F(u, v))} \right) \quad (3)$$

The wide dynamic range of the magnitude is compressed logarithmically, $L_c(u, v) = \ln(\max(M_c(u, v), \epsilon) + \epsilon)$, which makes the values better conditioned

for learning. The phase is periodic and jumps at the branch boundary, so it is encoded through its sine and cosine:

$$P_c^{\cos}(u, v) = \cos(\Phi_c(u, v)) \quad P_c^{\sin}(u, v) = \sin(\Phi_c(u, v)) \quad (4)$$

This removes the artificial discontinuities of raw phase while preserving its content. Formally, the raw phase $\Phi_c(u, v) \in (-\pi, \pi]$ lives on the circle S^1 , so representing it as a real scalar maps spectra whose true phases differ by an arbitrarily small amount near the branch cut to values separated by almost 2π ; a network consuming raw phase must therefore approximate a discontinuous function, and training becomes unstable precisely at the most informative frequencies. The mapping $\Phi \mapsto (\cos \Phi, \sin \Phi)$ is an injective, 1-Lipschitz embedding of S^1 into $\mathbb{R}^2$, since $\|(\cos \Phi_1, \sin \Phi_1) - (\cos \Phi_2, \sin \Phi_2)\|_2 = 2|\sin((\Phi_1 - \Phi_2)/2)| \leq |\Phi_1 - \Phi_2|$, with components bounded in $[-1, 1]$. Small phase perturbations, therefore, always produce proportionally small input changes, the property that makes phase learnable in practice.

The processed log-magnitude and phase data from three different channels are then combined into a single nine-channel frequency tensor

$$\mathbf{X}_{\text{freq}} \in \mathbb{R}^{9 \times H \times W}$$

This format keeps both the energy distribution and the spatial details, making it suitable for further processing with convolutional methods.

3.2 CNN architecture

To process the proposed nine-channel frequency representation, the backbone’s first convolutional layer is widened from three to nine input channels so the network can jointly learn from magnitude, cosine-phase, and sine-phase features; it maps the input to l feature channels with a 7×7 kernel, stride 2, and padding 3. To keep the benefits of ImageNet pretraining, a transfer-and-scale initialisation is used: the weights of the first three input channels are copied from the pretrained model, while the remaining six are set with scaled Kaiming initialisation.

$$W_{\text{conv1}}[:, 0:3, :, :] = W_{\text{pretrained}} \quad W_{\text{conv1}}[:, 3:9, :, :] = 0.1 \cdot W_{\text{Kaiming}} \quad (5)$$

which preserves useful low-level visual filters while letting the network adapt to spectral inputs. Feature extraction then follows the standard stages of the chosen backbone, and the final convolutional stage produces deep feature maps $\mathbf{A} \in \mathbb{R}^{B \times C_{\text{backbone}} \times h \times w}$, where B is the batch size, C_{backbone} the number of output channels and $h \times w$ the spatial resolution. These maps feed the dual-frequency feature fusion module and underpin the explainability mechanism.

3.3 Dual-frequency feature fusion

The proposed dual-frequency feature fusion (DFFF) module operates on deep activation maps $\mathbf{A} \in \mathbb{R}^{B \times C \times h \times w}$ that are taken from the last residual block.

To get more information about the spectral details in these spatial features, a second-level 2D FFT is done separately for each channel.

$$G_k(u, v) = \text{FFT2D}(A_k(x, y)), \quad M_k^{(2)} = |G_k|. \quad (6)$$

This gives magnitude spectra, which are then log-scaled and normalised for each channel to create frequency-domain feature maps $\mathbf{A}_{\text{freq}}$. This process reveals repeating structures, dominant spatial scales, and periodic patterns that aren't easily seen in the original spatial data, making the representation richer with more complex structural details.

The two streams are provably complementary. A convolutional activation A_k is translation-*equivariant*, i.e., shifting the input shifts the activation. Its magnitude spectrum, by the Fourier shift theorem, satisfies $|\mathcal{F}\{A_k(x - x_0, y - y_0)\}(u, v)| = M_k^{(2)}(u, v)$, i.e., it is translation-*invariant*, and by Parseval's theorem [8], it preserves the total feature energy while reorganising it by spatial scale. Every coefficient $M_k^{(2)}(u, v)$ aggregates evidence from the whole feature map, giving each fused unit a global receptive field that stacked local convolutions can only approximate with depth. The spatial stream thus encodes *where* structures occur, while the spectral stream encodes *which* periodicities and scales are present, and neither can be recovered from the other once the magnitude operation discards phase. It is this invariance-equivariance pairing with adaptive reweighting, rather than a mere concatenation of branches, that constitutes the core methodological contribution.

Because the two feature types occupy different numerical ranges, separate batch normalisation layers align their distributions before concatenation along the channel axis:

$$\mathbf{C} = [\text{BN}_{\text{spatial}}(\mathbf{A}); \text{BN}_{\text{freq}}(\mathbf{A}_{\text{freq}})] \in \mathbb{R}^{B \times 2C \times h \times w} \quad (7)$$

A channel attention mechanism based on global average pooling then reweights the joint tensor,

$$\mathbf{z} = \text{GAP}(\mathbf{C}), \quad \mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})), \quad \hat{\mathbf{C}} = \mathbf{s} \odot \mathbf{C} \quad (8)$$

where δ is ReLU, σ is the sigmoid, and W_1, W_2 are learnable projections. Because $\mathbf{s}$ is computed from the pooled statistics of both streams, the module can suppress spectral channels dominated by phase-induced noise and amplify those carrying stable structure on a per-input basis. Finally, $\hat{\mathbf{C}}$ is split back into its spatial part $\hat{\mathbf{A}}_s = \hat{\mathbf{C}}[:, : C]$ and frequency part $\hat{\mathbf{A}}_f = \hat{\mathbf{C}}[:, C :]$, which are merged through a learnable fusion:

$$\mathbf{A}_{\text{fused}} = \alpha \cdot \hat{\mathbf{A}}_s + \beta \cdot \hat{\mathbf{A}}_f \quad (9)$$

where $\alpha = \sigma(a)$ and $\beta = \sigma(b)$ are fusion weights constrained to $(0, 1)$ with trainable scalars a and b , letting optimisation itself determine the reliability ratio of the two domains rather than a hand-tuned constant. The fused feature combines local spatial detail and global spectral structure before pooling and classification; the ablation in Section 4 isolates its effect.

3.4 Classification Head

Global average pooling condenses $\mathbf{A}_{\text{fused}}$ into a fixed-length vector

$$\mathbf{v} = \frac{1}{h w} \sum_{i=1}^h \sum_{j=1}^w \mathbf{A}_{\text{fused}}(:, :, i, j) \quad (10)$$

which reduces sensitivity to spatial shifts. This vector passes through a three-layer fully connected head in which each stage applies dropout, a linear projection, ReLU, and batch normalisation, producing logits $\mathbf{o} \in \mathbb{R}^{B \times K}$ for the K classes. The dropout rates decrease across the stages ($p_1 > p_2 > p_3$), applying stronger regularisation early and lighter regularisation late, which curbs overfitting without discarding the detail needed for accurate predictions.

3.5 Explainability

Class activation mapping (CAM) has evolved into several variants, including Grad-CAM [32], XGrad-CAM [10], Smooth Grad-CAM [23], Score-CAM [41], Ablation-CAM [27], Eigen-CAM [21] and Grad-CAM++ [4]. Here, Score-CAM is adapted to the fused spectral-spatial representations from the DFFF module, though the procedure extends to other CAM methods. Working on the pre-pooling maps $\mathbf{A}_{\text{fused}} \in \mathbb{R}^{1 \times C \times h \times w}$, each channel is bilinearly upsampled to the input size and min-max normalised.

$$\bar{A}_k = \frac{A_k^\uparrow - \min(A_k^\uparrow)}{\max(A_k^\uparrow) - \min(A_k^\uparrow)} \quad (11)$$

where $A_k^\uparrow$ denotes the upsampled activation map obtained through bilinear interpolation. $\bar{A}_k$ serves as a soft spatial mask applied to the frequency-domain input, $\mathbf{X}_k = \mathbf{X}_{\text{freq}} \odot \bar{A}_k$, which retains the regions most relevant to the k -th fused feature. Each masked input is passed through the full network to obtain a confidence score $s_k = p_t(\mathbf{X}_k)$, the softmax probability of the target class t . The class activation map then weights the fused feature maps by these scores and keeps positive contributions:

$$\text{CAM}(u, v) = \text{ReLU} \left(\sum_{k=1}^C s_k \cdot A_{\text{fused}, k}(u, v) \right) \quad (12)$$

The resulting saliency map preserves the model-agnostic nature of Score-CAM while explicitly leveraging the fused spectral-spatial representations learned by the proposed DFFF module. Consequently, the generated explanations reflect both spatial localisation cues and discriminative frequency-domain characteristics that contribute to the final prediction.

3.6 Frequency to spatial domain mapping

To make the frequency-domain CAM humanly interpretable, it is used as a mask on the spectral magnitude and, together with the original phase, passed through the inverse FFT to yield a spatial saliency signal:

$$I_{\text{sal}} = \text{IFFT2D}(\text{CAM} \cdot M \cdot e^{j\Phi}) \quad (13)$$

The channels are averaged, and the map is inverted, since high spectral energy corresponds to edges and textures rather than object centres. The map is then smoothed with a Gaussian filter ($\sigma = 2.5$), thresholded at the 40th percentile, cleaned by morphological closing (5×5) followed by opening (3×3), and contrast-enhanced with a power-law transform (exponent 0.7). For image reconstruction, the encoding steps are simply reversed: the complex spectrum is rebuilt as $\hat{F}(u, v) = e^{\text{clamp}(L, -10, 10)} \cdot e^{j\Phi(u, v)}$ from the log-magnitude L and phase Φ , then de-centred with inverse `fftshift`, and the spatial image is recovered as the real part of its inverse FFT.

4 Experimentation and Results

4.1 Datasets used

This paper uses two publicly available datasets: Fruits-360 dataset and Brain Tumor MRI dataset. The Fruits-360 dataset is a large-scale image classification dataset available on Kaggle. It consists of RGB images of fruits and vegetables, each of size 100×100 pixels. The dataset contains a total of 130,344 images, divided into 110,792 training samples, 19,552 validation samples, and a separate test set of 43442 images. The Brain Tumor MRI dataset consists of labelled MRI scans used for multi-class brain tumor classification, including categories such as glioma, meningioma, and pituitary tumors. It contains a total of 5,600 images in the training set, which is further divided into 4,760 training samples and 840 validation samples, along with a separate test set of 1,600 images.

4.2 Training details

Training images pass through a stochastic augmentation pipeline: random horizontal flip ($p = 0.5$), vertical flip ($p = 0.3$), rotation within $\pm 20^\circ$, colour jitter on brightness/contrast/saturation (± 0.2) and hue (± 0.1), and random affine transforms with up to 10% translation and 0.9 to 1.1 scaling before conversion to $[0, 1]$ tensors and normalisation with ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$); the test set skips augmentation and receives only tensor conversion, the same normalisation, and FFT-based frequency conversion.

The objective is cross-entropy with label smoothing, $\mathcal{L} = -\sum_{k=1}^K \tilde{y}_k \log(\hat{y}_k)$ with $\tilde{y}_k = (1 - \varepsilon) y_k + \frac{\varepsilon}{K}$, $\hat{\mathbf{y}} = \text{softmax}(\mathbf{o})$, and $\varepsilon = 0.1$, which discourages over-confidence and aids generalisation. Optimisation uses AdamW with decoupled weight decay $\lambda = 5 \times 10^{-4}$; pretrained layers are fine-tuned at $\eta_{\text{pre}} = 10^{-5}$ while

the new components (modified input convolution, DFFF module, and classification head) use $\eta_{\text{new}} = 5 \times 10^{-4}$, under cosine annealing with warm restarts ($T_0 = 10$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 10^{-7}$) whose periodic restarts help escape local minima.

Automatic mixed precision computes in float16 with float32 parameter updates and dynamic gradient scaling, and since FFT operations can inflate gradients, the global gradient norm is clipped at 0.5. Experiments are run on a CUDA GPU (16 GB preferred) with PyTorch, torchvision, SciPy for saliency post-processing, and Matplotlib for visualisation; cuDNN benchmark mode is enabled and deterministic mode disabled for speed. Data are split 85/15 into training/validation (seed = 42), and models are trained for 50 epochs with batch size 128 (pinned memory and prefetch factor 2), dropout in the classifier, and early stopping after 15 epochs without validation improvement.

Table 1: Performance of the proposed spectral framework on the Fruits-360 test set. Each column corresponds to the proposed method instantiated with a different CNN backbone.

Metric	Proposed (ResNet-50)	Proposed (MobileNet V2)	Proposed (DenseNet-121)	Proposed (EfficientNet-B0)	Proposed (VGG-19)
Overall Accuracy (%)	90.37	89.94	89.65	87.78	86.51
Cohen’s Kappa Score (%)	90.34	89.88	89.60	87.71	86.45
Overall Precision (%)	92.25	90.75	91.54	91.35	89.27
Overall Recall (%)	90.48	89.89	89.90	87.82	86.71
Overall F1 Score (%)	90.30	89.73	89.79	87.72	86.40
Overall Specificity (%)	99.96	99.94	99.95	99.93	99.94
Overall Error Rate (%)	9.52	0.10	0.08	16.18	14.48

4.3 Quantitative results analysis

Table 1 and Table 2 presents the quantitative performance of the proposed dual-frequency feature fusion framework evaluated across five CNN backbone architectures: ResNet-50 [14], MobileNetV2 [30], DenseNet-121 [15], EfficientNet-B0 [34], and VGG-19 [33], on the Fruits-360 and Brain Tumor MRI datasets, respectively. The selected architectures cover a wide range of designs from deep and dense networks to lightweight and scaled models, ensuring robust validation of the framework across different model types and capacities.

These results show the benefit of this approach compared to pure spatial domain CNNs that only use pixel values and have trouble recognising repeating patterns, textures, and edges. The new method solves this by using a 9-channel frequency format based on FFT, which includes both the strength and the timing of the signals, giving extra information that goes beyond standard RGB inputs. The consistently high specificity and precision across both datasets and all five architectures indicate that the dual-frequency input reduces model confusion in

regions where spatial features are ambiguous, a common limitation of spatial-domain CNNs in fine-grained and medical image classification.

Table 2: Performance of the proposed spectral framework on the Brain Tumor MRI test set. Each column corresponds to the proposed method instantiated with a different CNN backbone.

Metric	Proposed (ResNet-50)	Proposed (MobileNetV2)	Proposed (DenseNet-121)	Proposed (EfficientNet-B0)	Proposed (VGG-19)
Overall Accuracy (%)	85.44	86.78	87.62	83.51	86.09
Cohen’s Kappa Score (%)	82.65	85.10	85.93	81.74	83.91
Overall Precision (%)	87.32	88.75	89.25	86.35	87.89
Overall Recall (%)	85.57	86.46	87.81	85.43	86.28
Overall F1 Score (%)	85.50	86.37	87.73	85.26	86.14
Overall Specificity (%)	95.86	96.04	97.05	94.59	95.97
Overall Error Rate (%)	14.27	12.26	11.08	16.38	13.05

4.4 Ablation study

An ablation study is conducted with the best backbone per dataset (ResNet-50 for Fruits-360 and DenseNet-121 for Brain Tumor MRI). Starting from the plain spatial-domain backbone, three components are enabled one at a time: (i) the 9-channel FFT input, (ii) sine–cosine phase encoding, and (iii) the dual-frequency feature fusion (DFFF) module with channel attention. Table 3 summarises the results. Three observations emerge. First, the magnitude spectrum alone improves over the spatial baseline, confirming that spectral energy carries discriminative information not readily accessible from raw pixels. Second, sine–cosine phase encoding stabilises the otherwise perturbation-sensitive phase and adds localisation capability. Third, the DFFF module yields the largest single improvement, showing that re-examining deep activations in the frequency domain and adaptively reweighting the two streams is essential. The full configuration outperforms every reduced variant on both datasets.

4.5 Qualitative results analysis

Figure 3 illustrates the activation maps generated by eight different Class Activation Mapping (CAM) variants applied to a cantaloupe sample from both datasets. Each variant offers distinct visualisation characteristics that reveal how the dual-frequency feature fusion model interprets frequency-enriched features during classification. Spatial-domain CAM visualisations frequently produce diffuse, poorly localised heatmaps that spread activation across broad image regions rather than clinically meaningful structures, making them difficult to interpret and less useful for medical decision support.

Table 3: Ablation study of the proposed framework. Each row builds upon the previous configuration by adding one component. Results are reported as test-set accuracy (%).

Configuration	Fruits-360	Brain Tumor MRI
Spatial baseline (RGB input only)	87.98	85.08
+ FFT magnitude input (3 channels)	88.75	85.66
+ Sine-cosine phase encoding (9 channels)	89.18	86.23
+ DFFF module with channel attention	90.37	87.62

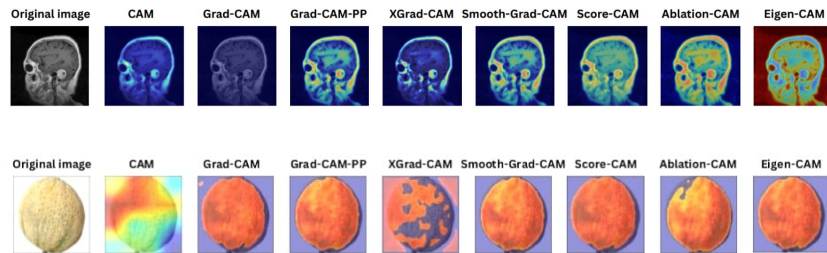


Fig. 3: Qualitative comparison of frequency-domain heatmaps with class activation mapping methods.

The proposed spectral learning framework addresses this limitation by providing structured frequency-domain information, enabling activation maps to align with real anatomical and pathological features. Its precise localisation comes from using magnitude, cosine phase, and sine phase components, which capture high-frequency boundary and texture details often suppressed in spatial representations.

5 Conclusion and Future Work

This paper introduced an explainable spectral learning framework in which CNNs operate directly on a 9-channel frequency-domain representation built from the log-magnitude, cosine-phase, and sine-phase of each RGB channel. The sine-cosine formulation keeps phase information continuous and therefore stable to learn from, while the dual-frequency feature fusion module, which re-examines the deepest activations through a second-level FFT and adaptively reweights the spatial and spectral streams via channel attention, compensates for the noise that explicit phase encoding can introduce. Crucially, the framework closes the interpretability gap of frequency-domain models: CAM-style attributions are computed on the fused spectral features and projected back to the pixel space through the inverse FFT so that predictions made from spectra can still be verified visually in the original image. Experiments across five CNN backbones on

the Fruits-360 and Brain Tumor MRI datasets showed consistent classification performance (up to 90.37% and 87.62% accuracy, respectively) together with saliency maps that align with meaningful objects and anatomical structures, and the ablation study confirmed that the input encoding, phase representation, and fusion module each contribute complementary gains.

The present study also has limitations that suggest concrete directions for future work. The evaluation covers two datasets and CNN backbones only; validating the framework on larger clinical cohorts, additional modalities such as CT and histopathology, and transformer-based architectures is a natural next step. Building on the current design, a multi-resolution frequency pyramid (100×100 , 50×50 , 25×25) with lightweight CNN processing and attention-based fusion will be developed to capture spectral structure at several scales, accompanied by a multi-scale CAM formulation for frequency-aware interpretability.

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